

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering
B. Sc. Engineering 1st Year 2nd Term Examination, 2020
EE 1213

(Electrical Circuits and Electronics)

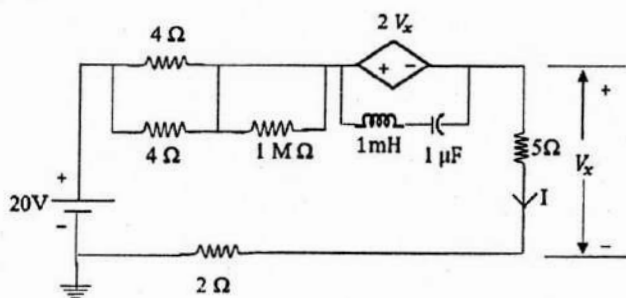
Time: 1 Hour 30 Minutes.

Full Marks: 120

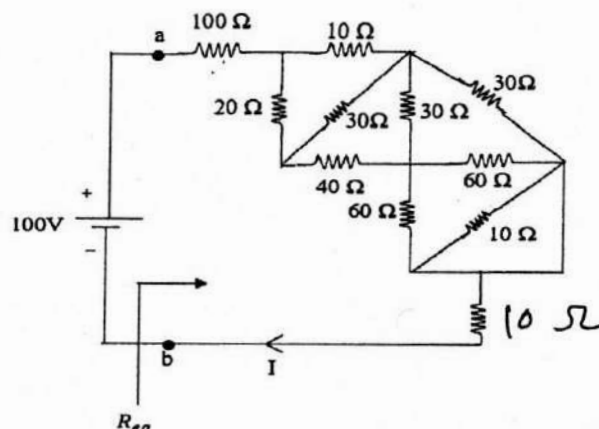
- N.B. i) Answer any two questions from each section.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

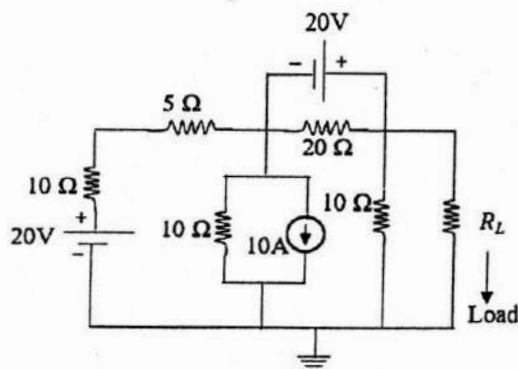
- 1(a). Differentiate between i) Active elements and passive elements. 06
ii) Linear and Non-linear network
- 1(b). Determine the current I flowing through $5\ \Omega$ resistor and all the node voltages for the circuit shown in figure below. 10



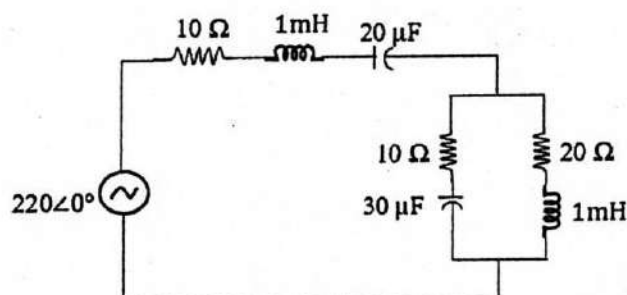
- 1(c). Determine the equivalent resistance R_{eq} seen from terminal ab and the current I for the circuit shown in figure. 14



- 2(a). State and explain maximum power transfer theorem. Show that, maximum power received by the load is 50% of the supplied power. 14
- 2(b). Mention the conditions to apply superposition theorem. Determine the Thevenin equivalent circuit with respect to load R_L for the circuit shown in figure. 16



- 3(a). What is power factor (P.F)? How power factor can be improved ? 07
- 3(b). Derive the expression of impedance Z for a series RLC circuit hence show that power factor is unity for the RLC series circuit at resonance. 10
- 3(c). For the circuit shown in figure, determine the (i) impedance, (ii) branch currents , (iii) voltages across each branch, and (iv) power consumed by each elements. 13



SECTION - B

- 4(a). Define Fermi energy level. Show that "the Fermi energy level shifts due to n-type and p-type doping but remains exactly in the middle of the band gap for intrinsic semiconductor" 15
- 4(b). State and derive the mass action law, $n_i^2 = np$ for intrinsic semiconductor. 15
- 5(a). Write short notes on: i) Avalanche breakdown and ii) Zener breakdown 06
- 5(b). Show "Depletion region of a p-n junction diode at zero applied bias is proportional to the square root of built-in potential" (V_0). 12
- 5(c). For a Ge- semiconductor material, $N_C = 1.5 \times 10^{18} \text{ cm}^{-3}$, $N_V = 1.3 \times 10^{19} \text{ cm}^{-3}$ at $T = 300 \text{ K}$. 12
- i) Determine the intrinsic carrier concentration of the semiconductor at $T = 300 \text{ K}$.
- ii) Find the position of the intrinsic Fermi level with respect to the center of the band gap.
- 6(a). Draw the energy band diagram of p-n junction diode for the following conditions. 10
- i) at zero applied bias
- ii) at $V_F = \frac{A}{10} \text{ Volt}$, where $A =$ last two digit of your roll number.
- 6(b). Differentiate between Latch and Flip-Flop 06
- 6(c). Define synchronous and asynchronous counter. Design a BCD counter using T-Flip- Flop. 14

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 2nd Term Examination, 2020

Hum 1213

(Technical English)

Time: 1 Hour 30 Minutes

Full Marks: 120

- N.B. i) Answer any TWO questions from each section.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a) Write one synonym for each of the following and use the new words in sentences. 10
Elegant, Captivate, Fatal, Hazard
- 1(b) Transform the following sentences as instructed below: 10
i. Nazrul was the most brilliant poet of his time. (Comparative)
ii. Old fools surpass all the other fools in folly. (Negative)
iii. Rafa owed his success to his father. (Complex)
iv. If you are not Bangladeshi you cannot be admitted. (Simple)
- 1(c) Make sentence with the following idioms and phrases. 10
Eat a humble pie, Bear down, By hook and by crook, The knives are out.
- 2(a) Write what you would say in the following situations. 15
i. Meeting Neena for the first time.
ii. Welcoming everyone to an international conference at KUET.
iii. Politely respond to an apology of you employee.
iv. Request your colleague for some assistance at work.
v. Politely decline a request to receive help.
- 2(b) Make sentences with each of the following structure. 15
i. Sub+Linking verb+to be + Adjective.
ii. Sub+Intransitive verb +Adjective Complement.
iii. There+Intransitive Verb+ Sub+ Prepositional Phrase.
iv. Sub+ Transitive Verb + Object.
v. Sub+ Intransitive Verb+ Infinite Phrase.
- 3(a) Complete the following sentences with subordinate clauses as directed within brackets. 10
i. Fatema did not tell us _____. (Noun Clause)
ii. There is a rumor _____. (Noun Clause)
iii. The match will be postponed _____. (Adverb Clause)
iv. The man _____ die. (Adjective Clause)
- 3(b) Make new word with each of the following prefixes and suffixes and use the new words 10
in sentences
Ab_____, Ultra_____, _____mony, _____ ate.
- 3(c) Change the following words as directed and make sentences with the new words 10
Soft (into verb), trust (into adj), useless (noun), provide (into adverb)

SECTION – B

4. Read the text carefully and answer the questions.

30

A man in the moving crowd heard his cry, and stooping with very great difficulty, lifted him in his arms. 'How did you get here, child? Whose baby are you?' the man asked as he moved away from the crowd. The child wept more bitterly than ever now and only cried, 'I want my mother, I want my father.' The man tried to soothe him by taking him up to the roundabout. 'Will you have a ride on the horses?' he gently asked as he approached the ring. The child's throat tore into a thousand shrill sobs and he only shouted, 'I want my mother, I want my father.' The man headed towards the place where the juggler still played on the flute to the dancing cobra. 'Listen to that music, child' he pleaded. But the child shut his ears with his finger and shouted even more loudly, 'I want my mother, I want my father'.

The man took him near the balloons, thinking the bright colours of the balloons would distract the child's attention. 'Would you like a rainbow-coloured balloon?' he asked persuasively. But the child turned his eyes from the balloons and just sobbed, 'I want my mother, I want my father'. Then, to make the child happy, the man took to the sweetmeat shop. 'What sweets would you like, child?' he asked. The child turned his face away from the sweetmeat shop and only sobbed, 'I want my mother, I want my father'.

- i. Who rescued the boy and how?
- ii. Why was the boy not consoled? Give your reasons.
- iii. How did the gentlemen try to console the boy?
- iv. The gentleman should have looked for the parents or reported to the police instead of consoling the baby. Do you agree? Give reasons for your answer.
- v. Give a title of the text.

5. Write a cause and effect paragraph on 'COVID-19 pandemic in Bangladesh'

30

- 6(a) Amplify the idea of the statement
"Every cloud has a silver lining"

15

- 6(b) You are applying for a job and you need a letter of reference. Now, write a letter to your former teacher asking for a letter of reference.

15

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 2nd Term Examination, 2020

ESE 1205

(Thermodynamics for Energy Engineering)

Time: 1 Hour 30 Minutes.

Full Marks: 120

N.B. i) Answer any Two questions from each section.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Define thermodynamic system and classify it. 06
- 1(b). Write down the phase rule and mention the meaning of each term. 06
- 1(c). Suppose (i) a mixture of liquid and gaseous water & (ii) a mixture of liquid and gaseous air. Among them, which one is pure substance? Explain. 06
- 1(d). “An energy transfer can be heat or work, depending on system selection” – explain the statement. 12
- 2(a). Suppose a system has only work interaction with its surroundings. Does the system contain any entropy? Explain. 06
- 2(b). Which thermodynamic properties are based on the zeroth, first and second law of thermodynamics? Write the mathematical forms of first law of thermodynamics for control volume and control mass. 09
- 2(c). A stationary mass of gas is compressed without friction from an initial state of 0.3 m^3 and 0.105 MPa to final state of 0.15 m^3 and 0.105 MPa . There is a transfer of 37.6 kJ of heat from the gas during the process. Calculate the change of internal energy during the process. 15
- 3(a). “The efficiency of an irreversible heat engine is always less than the efficiency of a reversible one operating between the same two reservoirs” – proof the statement. 10
- 3(b). What is dead state? Is it possible to generate exergy? If not then why? 05
- 3(c). A 200 m^3 rigid tank contains compressed air at 1 MPa and 300 K . Determine the work that can be obtained from this air if the environmental conditions are 100 kPa and 300 K . 15

SECTION – B

- 4(a). What is power cycle? How the efficiency of Carnot cycle varies with boiler and condenser pressure? 08
- 4(b). Why Carnot cycle is not used in practical? 04
- 4(c). An engine operating on an air standard diesel cycle, the compression ratio is 17 and at the beginning of isentropic compression, the temperature is 20°C and the pressure is 100 kPa . Heat is added until the temperature at the end of constant pressure is 1500°C . Calculate (i) the cutoff ratio, (ii) heat supplied per kg of air, and (iii) the cycle efficiency. 18

- 5(a). Write down the advantages of vapor absorption refrigeration system over vapor compression refrigeration system. 10
- 5(b). Consider a steam power plant operating on the ideal Rankine cycle. Steam enters the turbine at 3 MPa and 355°C and is condensed in the condenser at a pressure of 11 kPa. Determine the thermal efficiency of this power plant. Data required for this question is given in below table 5(i) and 5(ii). 20

Table 5(i): Saturation pressures and properties of water and steam.

Press.(P)	Temp.(t)	Specific volume		Specific enthalpy			Specific entropy		
		Sat. liquid v_f	Sat. vapor v_g	Sat. liquid h_f	Evap. h_{fg}	Sat. vapor h_g	Sat. liquid s_f	Evap. s_{fg}	Sat. vapor s_g
kPa	°C	m ³ /kg	m ³ /kg	kJ/kg	kJ/kg	kJ/kg	kJ/kg K	kJ/kg K	kJ/kg K
10	45.81	0.001010	14.674	191.83	2392.8	2584.7	0.6493	7.5009	8.1502
11	47.69	0.001011	13.415	199.67	2388.3	2588.0	0.6738	7.4430	8.1168
12	49.42	0.001012	12.361	206.92	2384.1	2591.1	0.6963	7.3900	8.0863

Table 5(ii): Superheated steam properties.

$P(t_{sat}) = 3 \text{ MPa (233.90°C)}$

t °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg K
340	0.08871	2825.5	3091.6	6.7045
350	0.09053	2843.7	3115.3	6.7428
360	0.09233	2861.7	3138.7	6.7801

- 6(a). State Dalton's law of partial pressure and Amagat's law. 06
- 6(b). What are meant by absolute humidity, relative humidity and degree of saturation? 06
- 6(c). A mixture of ideal gas consists of 3 kg of N₂ and 5 kg of CO₂ at a pressure of 300 kPa and a temperature of 20°C. Calculate: (i) mole fraction of each constituent, (ii) equivalent molecular weight of the mixture, (iii) equivalent gas constant of the mixture, (iv) partial pressure, (v) density of the mixture, and (vi) c_p , c_v of mixture. Assume γ for CO₂ and N₂ are 1.29 and 1.40 respectively. 18

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 2nd Term Examination, 2020

Math 1213

(Differential Equation & Coordinate Geometry)

Time: 1 hour 30 min.

Full Marks: 120

- N.B. i) Answer any two questions from each section.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

Section A

- 1(a). Identify the conic $x^2 + 2xy + y^2 - 6x - 2y + 4 = 0$ and from it, remove the xy term. 14
- 1(b). Determine the angle through which axes must be rotated to remove the xy term in the equation $7x^2 - 6\sqrt{3}xy + 13y^2 = 16$. Also find the transformation equation. 13
- 1(c). When does the general equation of second degree represents (i) a parabola, (ii) an ellips, and (iii) a hyperbola. 03
- 2(a). Find the cylindrical polar coordinates and spherical polar coordinates, whose cartesian coordinates is $4, -4, 4\sqrt{6}$. 12
- 2(b). Find the equation of the plane through the vectors $\hat{i} - \hat{j} + \hat{k}$ and $3\hat{i} + \hat{j} + 2\hat{k}$ and parallel to the vector $\hat{i} + 5\hat{k}$. 12
- 2(c). Find the equation of plane that passes through origine and perpendicular to the line joining the points $(4, 5, -1)$ and $(2, 3, 6)$. 06
- 3(a). Find the shortest distance between the lines $\frac{x}{4} = \frac{y+1}{3} = \frac{z-2}{2}$ and $5x - 2y - 3z + 6 = 0 = x - 3y + 2z - 3$. 20
- 3(b). Find the equation of a straight line which passes through a point $3\hat{i} - 2\hat{j} + 5\hat{k}$ and parallel to the vector $2\hat{i} + \hat{j} - 4\hat{k}$. 10

Section B

4(a). Define general solution and boundary value problem. 18

Solve $(x^2 + y^2 + y)dx - xdy = 0$.

4(b). If the temperature of the air is 30°C and the substance 12
cools from 100°C to 70°C in 15 minutes. When will the
temperature of the substance be 40°C ?

5(a). Determine the general solution of 15

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = xe^{2x} \sin(x)$$

5(b). Solve 15

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 3y = e^x + 9x^2 + 5$$

6(a). Define Laplace transform. Find the Laplace transform of 14

$$f(t) = \begin{cases} \sin(t), & 0 < t < \pi \\ 0, & \pi < t < 2\pi \end{cases}$$

6(b). State Convolution theorem. Find the inverse Laplace 16
transform of $\frac{6s-4}{s^2-4s+20}$

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 2nd Term Examination, 2020

Ch 1213

(Chemistry II)

Time: 1 Hour 30 Minutes.

Full Marks: 120

- N.B. i) Answer any Two questions from each section.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Prove that half-life of a 1st order reaction does not depend on initial concentration. 08
- 1(b). How did Arrhenious explain the temperature dependence of reaction rates? How do the energy of activation can be calculated from Arrhenious equation? 12
- 1(c). Discuss briefly the activated complex theory of reaction rates. 10
- 2(a). State Beer-Lambert's law and deduce a mathematical expression for the law. 12
- 2(b). Why the photochemical chain reaction of H_2 and Cl_2 are high quantum yield reaction? 10
- 2(c). One mole reactants give 10 moles of products in a photochemical reaction. Calculate the energy associated with the reaction. 08
- 3(a). Does absorption convey the same meaning as adsorption? If not, how do they differ? 07
- 3(b). Deduce a Langmuir adsorption isotherm equation. 13
- 3(c). How does a catalyst accelerate the speed of a reaction? Explain with peoper energy profile diagram. 10

SECTION – B

- 4(a). What are the distinguishing features of sp^3 and sp^2 hybrid orbitals? 08
- 4(b). Discuss the lock and key mechanism of enzyme catalyzed reaction. 12
- 4(c). Trans compound has higher melting point than cis compound – 10
why?
- 5(a). Discuss about the different types of electronic transitions occurring 12
in organic molecule with diagram.
- 5(b). Calculate the vibrational degrees of freedom of water. Justify their 08
IR activities.
- 5(c). Explain the terms chromophores and auxochromes. 10
- 6(a). Briefly describe the mechanism of electrochemical corrosion. 12
- 6(b). Explain the term: Galvanizing. 08
- 6(c). What is anodic protection of corrosion? 10

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